



CY 513: Chemical Kinetics and Reaction Dynamics

Course Code	CY 513
Course Name	Chemical Kinetics and Reaction Dynamics
L-T-P-C	3-0-0-3
Prerequisites	Undergraduate-Level Physical Chemistry or Teacher's Consent
Intended for	BS Chemical Sciences and MSc Chemistry
Distribution	Core
Mutual Exclusion	-
Approval	8th Senate (Oct 2015)
Revised	59th BoA (Apr 2025) - syllabus revised

Course Contents

- **Kinetic Theory and Chemical Reactions:** Introduction Review of the kinetic theory of gases, the kinetic model of gases, collisions with walls and surfaces, the rate of effusions, transport properties of a perfect gas, atomic and molecular collisions, collisional theory, diffusion-controlled reactions, thermodynamics properties of diffusion, potential energy surface, generation, interpretation and correlation with reaction energetics; elementary ideas on conical intersection. [12 Lectures]
- **Chemical Kinetics and Rate Theories:** The rates of chemical reactions, Reaction Rates and Rate Laws, Order and Molecularity of Reactions, Experimental Techniques for Kinetic Measurements, Temperature Dependence of Reaction Rates: Arrhenius Equation, Collision Theory and Molecular Interpretations, Transition State Theory (TST) and Activation Energy, Potential Energy Surfaces and Reaction Pathways, RRKM Theory and Unimolecular Reactions. [12 Lectures]
- **Reaction Dynamics and Molecular Interactions:** Energy Transfer and Collision Dynamics, Reactive Collisions, Elementary vs. Complex Reactions, Atom-diatom reactions, Polyatomic reactions, State-selective, Reaction rates and cross sections, Molecular Beam and Laser Spectroscopy Studies in Reaction Dynamics, Scattering - Classical and Quantum, The Role of Solvent Effects in Reaction Kinetics. [10 Lectures]
- **Catalysis – Principles and Mechanisms:** Homogeneous vs. Heterogeneous Catalysis, Enzyme Catalysis and Michaelis-Menten Kinetics, Adsorption Isotherms (Langmuir, BET), Catalytic Cycles and Reaction Mechanisms Catalytic Reactions, Reaction Energetics, Activation and Deactivation of Catalysts, Auto Catalysis and its Mechanism. [8 Lectures]

References: (Books and articles)

1. R. D. Levine, **Molecular Reaction Dynamics**, Cambridge University Press 2005.
2. Henriksen & Hansen, **Theories of Molecular Reaction Dynamics**, Oxford University Press.
3. K.J. Laidler, **Chemical Kinetics**, 3rd Edition, Pearson Books.
4. Donald A. McQuarrie and John D. Simon, **Physical Chemistry: A Molecular Approach**, Viva Books, 1998.
5. Peter Atkins and Julio de Paula, **Physical Chemistry**, 7th Edition, Oxford University Press, 2002.
6. R. Schinke, **Photodissociation Dynamics**, Cambridge University Press, 1993.